

Module 2 purpose: convert damage into electric field

Module 2 is the electrostatic bridge between radiation-induced defects and carrier transport.

$$\{C_i, z_i, n, p, N_D^+, N_A^-, \text{bias/BCs}\} \longrightarrow \rho(x, y, t) \longrightarrow \phi(x, y, t) \longrightarrow \mathbf{E}(x, y, t)$$

- ▶ The central unknown is the electrostatic potential $\phi(x, y, t)$ in a 2-D silicon domain Ω .
- ▶ Radiation defects enter through their concentrations $C_i(x, y, t)$ and effective charge states z_i .
- ▶ The electric field is not solved independently; it is recovered after the potential solve:

$$\mathbf{E}(x, y, t) = -\nabla\phi(x, y, t).$$

- ▶ Module 2 supplies ϕ and $\mathbf{E}$ to carrier drift-diffusion, recombination, degradation metrics, and later coupled multiphysics updates.

Physical interpretation: charged defects perturb the local space charge; space charge bends the electrostatic potential; the potential gradient drives carrier motion.

Governing electrostatics and defect space charge

Electrostatic limit of Maxwell's equations

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho, & \mathbf{D} &= \epsilon \mathbf{E}, & \mathbf{E} &= -\nabla \phi, \\ \Rightarrow \quad \nabla \cdot (\epsilon \nabla \phi) &= -\rho. \end{aligned}$$

Total semiconductor space charge

$$\rho(x, y, t) = q[p(x, y, t) - n(x, y, t) + N_D^+(x, y) - N_A^-(x, y)] + \rho_{\text{def}}(x, y, t),$$

$$\rho_{\text{def}}(x, y, t) = q \sum_{i=1}^M z_i C_i(x, y, t).$$

- ▶ p and ionized donors N_D^+ contribute positive charge; n and ionized acceptors N_A^- contribute negative charge.
- ▶ A defect species contributes positive, negative, or neutral charge according to the sign and magnitude of z_i .
- ▶ For spatially uniform silicon permittivity,

$$\epsilon = \epsilon_{\text{Si}}, \quad \epsilon_{\text{Si}} \nabla^2 \phi = -\rho.$$

- ▶ In nonuniform media or at material interfaces, the conservative form $\nabla \cdot (\epsilon \nabla \phi) = -\rho$ must be retained.

Finite-element weak form and Module 2 outputs

Weak form on the 2-D domain Ω

$$\int_{\Omega} \epsilon \nabla w \cdot \nabla \phi \, d\Omega = \int_{\Omega} w \rho \, d\Omega + \int_{\Gamma_N} w g_N \, d\Gamma$$

$$\phi_h(x, y, t) = \sum_{j=1}^N \Phi_j(t) N_j(x, y),$$

$$K_{ij} = \int_{\Omega} \epsilon \nabla N_i \cdot \nabla N_j \, d\Omega,$$

$$b_i(t) = \int_{\Omega} N_i \rho(x, y, t) \, d\Omega + \int_{\Gamma_N} N_i g_N \, d\Gamma,$$

$$K\Phi(t) = \mathbf{b}(t), \quad \mathbf{E}_h|_e(t) = -\nabla \phi_h|_e(t).$$

- ▶ Dirichlet boundaries impose contact voltages or a reference ground: $\phi = \phi_D$ on Γ_D .
- ▶ Neumann boundaries impose normal electric displacement or insulating/symmetry conditions: $-\epsilon \nabla \phi \cdot \hat{\mathbf{n}} = g_N$ on Γ_N .
- ▶ Main outputs: nodal potential Φ , reconstructed field $\mathbf{E}_h$, and residual diagnostics.
- ▶ Module 2 is quasi-static: at each coupled time level, the current charge density determines the instantaneous electrostatic solution.

Module 3 purpose: evolve the defect landscape

Module 3 converts the initial radiation damage map into a time-dependent defect concentration field.

Module 1 damage map $\longrightarrow C(x, y, 0)$ or $S(x, y, t) \longrightarrow C(x, y, t) \longrightarrow$ Modules 2, 4, and 5

Local defect balance law

$$\begin{aligned}\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J}_C &= R(C, T, \dots), \\ \mathbf{J}_C &= -D \nabla C, \\ \Rightarrow \quad \frac{\partial C}{\partial t} &= \nabla \cdot (D \nabla C) + R(C, T, \dots).\end{aligned}$$

- ▶ $C(x, y, t)$ is a coarse-grained concentration of radiation-induced defects in the 2-D silicon domain.
- ▶ $\mathbf{J}_C$ is the defect flux; Fick's law sends defects down the concentration gradient.
- ▶ R contains local creation, removal, recombination, clustering, dissociation, or transformation.
- ▶ For multiple defect species,

$$\frac{\partial C_i}{\partial t} = \nabla \cdot (D_i \nabla C_i) + R_i(C_1, C_2, \dots, T).$$

Diffusion, annealing, temperature dependence, and boundaries

Baseline single-species diffusion–reaction model

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right) + \frac{\partial}{\partial y} \left(D \frac{\partial C}{\partial y} \right) - k_{\text{ann}} C + S$$

Thermally activated coefficients

$$D(T) = D_0 \exp \left(-\frac{E_D}{k_B T} \right),$$

$$k_{\text{ann}}(T) = k_0 \exp \left(-\frac{E_{\text{ann}}}{k_B T} \right).$$

- ▶ D controls spatial migration; k_{ann} controls local defect removal or transformation.
- ▶ Higher T generally increases both migration and annealing by reducing the Boltzmann penalty for crossing activation barriers.
- ▶ Important timescales:

$$\tau_D \sim \frac{L^2}{D}, \quad \tau_{\text{ann}} \sim \frac{1}{k_{\text{ann}}}.$$

- ▶ If $\tau_D \ll \tau_{\text{ann}}$, defects redistribute before disappearing; if $\tau_D \gg \tau_{\text{ann}}$, they anneal locally before spreading.
- ▶ Default closed-domain boundary:

$$\hat{\mathbf{n}} \cdot \mathbf{J}_C = 0, \quad \hat{\mathbf{n}} \cdot D \nabla C = 0.$$

Finite-element form, update equation, and verification checks

Weak form for the 2-D defect-evolution equation

$$\int_{\Omega} w \frac{\partial C}{\partial t} d\Omega + \int_{\Omega} D \nabla w \cdot \nabla C d\Omega + \int_{\Omega} w k_{\text{ann}} C d\Omega = \int_{\Omega} w S d\Omega$$

$$C_h(x, y, t) = \sum_{j=1}^N C_j(t) N_j(x, y),$$

$$M_{ij} = \int_{\Omega} N_i N_j d\Omega,$$

$$(K_D)_{ij} = \int_{\Omega} D \nabla N_i \cdot \nabla N_j d\Omega,$$

$$(K_R)_{ij} = \int_{\Omega} k_{\text{ann}} N_i N_j d\Omega.$$

Backward-Euler update

$$[M + \Delta t(K_D + K_R)] \mathbf{C}^{m+1} = M \mathbf{C}^m + \Delta t \mathbf{s}^{m+1}$$

- ▶ Main output: $C(x, y, t)$, which feeds defect charge in Module 2 and defect-dependent coefficients in later modules.
- ▶ Pure annealing check: $D = 0 \Rightarrow C(t) = C(0)e^{-k_{\text{ann}} t}$.
- ▶ Pure diffusion with zero-flux boundaries: $R = 0 \Rightarrow M(t) = \int_{\Omega} C d\Omega$ should be conserved.
- ▶ Uniform initial state should remain uniform if sources and boundaries preserve uniformity.

Module 4 purpose: thermal response beyond pure Fourier diffusion

Module 4 computes the 2-D temperature or medium-energy field driven by radiation heating and later coupled electrical heating.

$$\{\dot{q}_{\text{rad}}, Q_J, Q_{\text{rec}}, Q_{\text{def}}, \text{thermal BCs}\} \longrightarrow T(x, y, t) \longrightarrow \{D_i, k_{\text{ann},i}, \mu_n, \mu_p, R\}$$

Conservation of thermal energy

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{q} = \dot{q}, \quad u = CT \quad \text{for constant volumetric heat capacity } C.$$

- ▶ $u(x, y, t)$ is internal energy density; $T(x, y, t)$ is the equivalent local temperature.
- ▶ $\dot{q}$ may represent thermalized radiation energy deposition, Joule heating, recombination heat, or defect-reconfiguration heat.
- ▶ Module 4 feeds temperature-dependent coefficients back to defect evolution, carrier transport, and coupled degradation metrics.
- ▶ The motivation for ballistic-diffusive physics is that localized, short-time heating can propagate with finite carrier flight time before reducing to ordinary diffusion.

Fourier, Cattaneo, and ballistic-diffusive thermal transport

Fourier conduction is the local diffusive limit.

$$\mathbf{q}_F = -k\nabla T,$$
$$C \frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + \dot{q}.$$

Cattaneo adds flux relaxation and finite thermal-wave speed.

$$\tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -k\nabla T,$$
$$\tau C \frac{\partial^2 T}{\partial t^2} + C \frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + \dot{q} + \tau \frac{\partial \dot{q}}{\partial t}.$$

- ▶ Fourier is accurate when the mean free path is much smaller than the device length.
- ▶ Cattaneo fixes the infinite-propagation-speed artifact but is still a local closure.
- ▶ Ballistic-diffusive modeling splits heat transport into an unscattered ballistic channel and a scattered medium channel:

$$u = u_b + u_m, \quad \mathbf{q} = \mathbf{q}_b + \mathbf{q}_m.$$

- ▶ The key dimensionless indicator is the Knudsen number $\text{Kn} = \Lambda/L$.

Module 4 target equation and coupling inputs

Reduced 2-D ballistic-diffusive medium-temperature equation

$$\tau C \frac{\partial^2 T_m}{\partial t^2} + C \frac{\partial T_m}{\partial t} = \nabla \cdot (k \nabla T_m) - \nabla \cdot \mathbf{q}_b + \left(\dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right)$$

$$\dot{q}_e = \dot{q}_{\text{rad}} + Q_J + Q_{\text{rec}} + Q_{\text{def}} + Q_{\text{ext}}.$$

- ▶ $\mathbf{q}_b$ is not a claim that the whole device is ballistic; it represents a retained finite-flight-time correction.
- ▶ In the diffusive limit, $\tau \rightarrow 0$ and $\mathbf{q}_b \rightarrow 0$, the equation reduces to the ordinary heat equation.
- ▶ Temperature modifies Module 3 defect kinetics:

$$D_i(T) = D_{i0} \exp\left(-\frac{E_{D,i}}{k_B T}\right), \quad k_{\text{ann},i}(T) = k_{i0} \exp\left(-\frac{E_{\text{ann},i}}{k_B T}\right).$$

- ▶ Temperature also modifies carrier mobilities, recombination rates, and later the strength of Joule-heating feedback.

Module 4 finite-element update and verification logic

Simplified Fourier-limit FEM form used as a baseline thermal path

$$\int_{\Omega} w C \frac{\partial T}{\partial t} d\Omega + \int_{\Omega} k \nabla w \cdot \nabla T d\Omega = \int_{\Omega} w \dot{q} d\Omega + \int_{\Gamma_N} w q_N d\Gamma.$$

$$T_h = \sum_j \Theta_j N_j,$$

$$M_{ij}^T = \int_{\Omega} C N_i N_j d\Omega,$$

$$K_{ij}^T = \int_{\Omega} k \nabla N_i \cdot \nabla N_j d\Omega,$$

$$\left[M^T + \Delta t K^T \right] \Theta^{m+1} = M^T \Theta^m + \Delta t \mathbf{f}_T^{m+1}.$$

- ▶ Dirichlet boundaries impose fixed temperature; Neumann boundaries impose heat flux; Robin boundaries impose convective exchange.
- ▶ The full Module 4 target keeps ballistic-memory terms, but the Fourier-limit solve is the necessary first verification baseline.
- ▶ Verification checks: uniform equilibrium preservation, diffusive-limit recovery, localized-pulse spreading, and sensitivity to finite-flight-time corrections.
- ▶ Main output: $T(x, y, t)$ on the same 2-D domain used by Modules 2, 3, 5, and 6.

Module 5 purpose: carrier transport in the damaged device

Module 5 evolves electron and hole densities under electric fields, diffusion, generation, recombination, and defect-modified material properties.

$$\{\phi, \mathbf{E}, C_i, T, G, R, \mu_n, \mu_p, D_n, D_p\} \longrightarrow n(x, y, t), p(x, y, t), \mathbf{J}_n, \mathbf{J}_p$$

Particle balance laws

$$\begin{aligned}\frac{\partial n}{\partial t} &= \frac{1}{q} \nabla \cdot \mathbf{J}_n + G - R, \\ \frac{\partial p}{\partial t} &= -\frac{1}{q} \nabla \cdot \mathbf{J}_p + G - R.\end{aligned}$$

- ▶ Electrons and holes have opposite charge, so their current-divergence signs differ in the continuity equations.
- ▶ G creates electron-hole pairs; R removes them through recombination.
- ▶ The electric field comes from Module 2, while defect and temperature fields from Modules 3–4 modify the transport coefficients.
- ▶ Module 5 produces carrier fields that feed back into the Poisson charge density in Module 6.

Drift-diffusion currents and 2-D carrier PDEs

Current-density closures

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n,$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p.$$

Drift-diffusion PDE system

$$\frac{\partial n}{\partial t} = \nabla \cdot (\mu_n n \mathbf{E} + D_n \nabla n) + G - R,$$

$$\frac{\partial p}{\partial t} = -\nabla \cdot (\mu_p p \mathbf{E} - D_p \nabla p) + G - R.$$

- ▶ Drift terms scale with mobility, density, and electric field.
- ▶ Diffusion terms respond to carrier-density gradients.
- ▶ In nondegenerate thermal equilibrium, the Einstein relation gives

$$D_n = \mu_n \frac{k_B T}{q}, \quad D_p = \mu_p \frac{k_B T}{q}.$$

- ▶ Steady-state current response is obtained by setting $\partial n / \partial t = \partial p / \partial t = 0$.

How defects degrade mobility and enhance recombination

Mobility degradation through additional momentum scattering

$$\begin{aligned}\mu_n &= \frac{q\tau_{m,n}}{m_n^*}, & \frac{1}{\tau_{m,n}} &= \frac{1}{\tau_{m,n}^{\text{lat}}(T)} + \frac{1}{\tau_{m,n}^{\text{dop}}} + \\ & & &+ \sum_j \sigma_{m,n}^{(j)} v_{\text{th},n} C_j, \\ \mu_p &= \frac{q\tau_{m,p}}{m_p^*}, & \frac{1}{\tau_{m,p}} &= \frac{1}{\tau_{m,p}^{\text{lat}}(T)} + \frac{1}{\tau_{m,p}^{\text{dop}}} + \sum_j \sigma_{m,p}^{(j)} v_{\text{th},p} C_j.\end{aligned}$$

Trap-assisted Shockley–Read–Hall recombination

$$R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad \frac{1}{\tau_n} = \sum_j \sigma_{n,j} v_{\text{th}} C_j, \quad \frac{1}{\tau_p} = \sum_j \sigma_{p,j} v_{\text{th}} C_j.$$

- ▶ Increasing defect concentration generally increases scattering and lowers mobility.
- ▶ Defects also act as trap populations, shortening carrier lifetime and increasing nonradiative recombination.
- ▶ These closures are the physical path from radiation damage to degraded current response.

Module 5 boundaries, FEM form, and outputs

Representative boundary conditions

Ohmic contacts: $n = n_c, \quad p = p_c,$

Insulating boundaries: $\mathbf{J}_n \cdot \hat{\mathbf{n}} = 0, \quad \mathbf{J}_p \cdot \hat{\mathbf{n}} = 0,$

Surface recombination: $\frac{1}{q} \mathbf{J}_n \cdot \hat{\mathbf{n}} = S_n(n - n_{\text{eq}}), \quad -\frac{1}{q} \mathbf{J}_p \cdot \hat{\mathbf{n}} = S_p(p - p_{\text{eq}}).$

Electron weak-form structure

$$\int_{\Omega} w \frac{\partial n}{\partial t} d\Omega + \int_{\Omega} \nabla w \cdot (\mu_n n \mathbf{E} + D_n \nabla n) d\Omega = \int_{\Omega} w (G - R) d\Omega + \text{boundary terms}.$$

- ▶ The hole weak form is analogous but carries the opposite drift-diffusion sign convention.
- ▶ Module 5 is nonlinear because current terms contain products such as $\mu_n n \mathbf{E}$ and recombination depends on n and p .
- ▶ Main outputs: $n, p, \mathbf{J}_n, \mathbf{J}_p$, recombination map, generation map, and current-voltage or degradation metrics.

Module 6 purpose: self-consistent multiphysics integration

Module 6 couples the fields from Modules 2–5 on the same 2-D device cross section.

$$\{C_j\} \leftrightarrow \rho \leftrightarrow \phi, \mathbf{E} \leftrightarrow n, p, \mathbf{J}_n, \mathbf{J}_p \leftrightarrow T \leftrightarrow \{C_j\}$$

- ▶ Module 2 supplies electrostatic potential and electric field from total charge density.
- ▶ Module 3 supplies defect concentration maps and defect-dependent charge, traps, and kinetic states.
- ▶ Module 4 supplies temperature and heat-transport response.
- ▶ Module 5 supplies carrier densities, currents, generation, recombination, and current-related degradation outputs.
- ▶ The goal is not four separate simulations, but a mutually consistent set of physical maps at each time level.

Coupled governing equations in Module 6

A compact coupled PDE system

$$\begin{aligned}\frac{\partial C_j}{\partial t} &= \nabla \cdot (D_j(T) \nabla C_j) + R_j(\{C\}, n, p, T) + S_j, \\ \nabla \cdot (\epsilon \nabla \phi) &= -q \left[p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right], \\ C_T \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + Q_{\text{rad}} + Q_J + Q_{\text{rec}} + Q_{\text{def}}, \\ \frac{\partial n}{\partial t} &= \nabla \cdot (\mu_n n \mathbf{E} + D_n \nabla n) + G - R, \\ \frac{\partial p}{\partial t} &= -\nabla \cdot (\mu_p p \mathbf{E} - D_p \nabla p) + G - R, \\ \mathbf{E} &= -\nabla \phi.\end{aligned}$$

- ▶ The equations are coupled through coefficients, sources, charge density, electric field, and heat generation.
- ▶ The thermal equation shown here is the compact diffusive version; Module 4 may replace it with the reduced ballistic-diffusive form.

Main physical feedback loops and staged coupling levels

Key feedback mechanisms

$$\begin{aligned}\{C_j, z_j\} &\rightarrow \rho_{\text{def}} \rightarrow \phi, \mathbf{E}, \\ \mathbf{E} &\rightarrow \mathbf{J}_n, \mathbf{J}_p \rightarrow n, p, \\ C_j &\rightarrow \mu_n, \mu_p, \tau_n, \tau_p \rightarrow \mathbf{J}_n, \mathbf{J}_p, R, \\ T &\rightarrow D_j, k_{\text{ann},j}, \mu_n, \mu_p, R, \\ \mathbf{J} \cdot \mathbf{E} &\rightarrow Q_J \rightarrow T.\end{aligned}$$

Recommended physical-coupling ladder

1. Level 0: one-way field transfer among modules.
2. Level 1: electrostatic-carrier feedback, $(n, p) \leftrightarrow \rho \leftrightarrow \phi, \mathbf{E}$.
3. Level 2: defect-electrostatic-carrier coupling, $C_j \rightarrow \rho_{\text{def}}, N_t \rightarrow R_{\text{SRH}}$.
4. Level 3: thermal feedback through T -dependent coefficients and current-induced heat.
5. Level 4: full self-consistent iteration among defects, electrostatics, carriers, and temperature.

Module 6 consistency checks and device-response outputs

Charge consistency

$$\rho_{\text{check}} = q \left[p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right], \quad \epsilon_\rho = \frac{\|\rho_{\text{Poisson}} - \rho_{\text{check}}\|_2}{\|\rho_{\text{check}}\|_2 + \rho_{\text{floor}}}.$$

Energy-source bookkeeping

$$Q_{\text{tot}} = Q_{\text{rad}} + Q_J + Q_{\text{rec}} + Q_{\text{def}} + Q_{\text{ext}}, \quad P_a(t) = \int_{\Omega} Q_a(x, y, t) d\Omega.$$

Carrier conservation in an insulating domain

$$\frac{d}{dt} \int_{\Omega} n d\Omega = \int_{\Omega} (G - R) d\Omega, \quad \frac{d}{dt} \int_{\Omega} p d\Omega = \int_{\Omega} (G - R) d\Omega.$$

- ▶ These diagnostics catch stale-field coupling, unit mistakes, and sign errors that can look numerically stable but be physically wrong.
- ▶ Device-level outputs may include current, leakage, recombination power, temperature rise, defect inventory, and degradation scores.

Module 2 PINN: surrogate mapping for Poisson electrostatics

Goal: learn the Module 2 input-output map while penalizing violations of electrostatics.

$$\mathcal{G}_\theta : (\mathbf{x}, \boldsymbol{\mu}) \longmapsto \phi_\theta(\mathbf{x}; \boldsymbol{\mu}), \quad \mathbf{E}_\theta = -\nabla_{\mathbf{x}} \phi_\theta$$

- ▶ $\mathbf{x} = (x, y)$ is the spatial coordinate; $\boldsymbol{\mu}$ contains problem parameters such as bias, defect amplitude, material constants, geometry, and source descriptors.
- ▶ The finite-element solver remains the reference model; the PINN is a fast differentiable surrogate for repeated Module 2 solves.
- ▶ A purely supervised neural network learns from pairs

$$(\rho, \text{BCs}, \boldsymbol{\mu}) \mapsto \phi_{\text{FEM}}.$$

- ▶ A physics-informed neural network also sees the governing equation and boundary conditions through its loss function.

Central distinction: the network is not a new electrostatic law; it is an approximate representation constrained to behave like the Poisson model.

Module 2 PINN: Poisson residual and boundary losses

Strong-form residual used at interior collocation points

$$\mathcal{R}_P(\mathbf{x}; \theta) = \nabla \cdot (\epsilon \nabla \phi_\theta(\mathbf{x})) + \rho(\mathbf{x})$$

For uniform silicon permittivity,

$$\mathcal{R}_P(\mathbf{x}; \theta) = \epsilon_{\text{Si}} \left(\frac{\partial^2 \phi_\theta}{\partial x^2} + \frac{\partial^2 \phi_\theta}{\partial y^2} \right) + \rho(\mathbf{x}).$$

Representative loss terms

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_r} \sum_{k=1}^{N_r} |\mathcal{R}_P(\mathbf{x}_k; \theta)|^2,$$

$$\mathcal{L}_D = \frac{1}{N_D} \sum_{k=1}^{N_D} |\phi_\theta(\mathbf{x}_k) - \phi_D(\mathbf{x}_k)|^2,$$

$$\mathcal{L}_N = \frac{1}{N_N} \sum_{k=1}^{N_N} |-\epsilon \nabla \phi_\theta(\mathbf{x}_k) \cdot \hat{\mathbf{n}} - g_N(\mathbf{x}_k)|^2.$$

Module 2 PINN: training objective and data construction

Combined supervised-plus-physics objective

$$\mathcal{L}_{\text{total}} = \lambda_s \mathcal{L}_{\text{data}} + \lambda_r \mathcal{L}_{\text{PDE}} + \lambda_D \mathcal{L}_D + \lambda_N \mathcal{L}_N + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$$

- Supervised samples come from trusted FEM cases:

$$\mathcal{D}_s = \{(\mathbf{x}_k, \boldsymbol{\mu}_k, \phi_{\text{FEM},k})\}_{k=1}^{N_s}.$$

- Interior residual points do not need FEM labels; they train the model to satisfy Poisson's equation between labeled samples.
- Boundary points enforce contact voltages, ground reference, insulating conditions, or prescribed normal flux.
- Inputs should be nondimensionalized, for example

$$\hat{x} = x/L, \quad \hat{\phi} = \phi/\phi_0, \quad \hat{\rho} = \rho/\rho_0.$$

- The loss weights must be scaled so that the largest-magnitude term does not dominate training for numerical rather than physical reasons.

Module 2 PINN: verification, strengths, and limits

A useful PINN surrogate must be judged against the reference electrostatic solve, not only by low training loss.

- ▶ Compare potential and field errors:

$$e_\phi = \frac{\|\phi_\theta - \phi_{\text{FEM}}\|_2}{\|\phi_{\text{FEM}}\|_2 + \phi_{\text{floor}}}, \quad e_E = \frac{\|\mathbf{E}_\theta - \mathbf{E}_{\text{FEM}}\|_2}{\|\mathbf{E}_{\text{FEM}}\|_2 + E_{\text{floor}}}.$$

- ▶ Check physically simple cases: zero charge, uniform space charge, linear potential, localized defect charge, and boundary-condition consistency.
- ▶ Strength: once trained, the surrogate may accelerate parameter sweeps and sensitivity scans.
- ▶ Limitation: a PINN can still satisfy a weakly weighted residual while violating charge conservation, boundary behavior, or sharp field gradients.
- ▶ Practical conclusion: keep FEM as the truth model; use the PINN only after systematic out-of-distribution testing.

Module 2 CPINN: why a causal training curriculum is needed

Module 2 is elliptic and quasi-static, so causality does not mean electrostatic time evolution.

$$\nabla \cdot (\epsilon \nabla \phi) = -\rho, \quad \mathbf{E} = -\nabla \phi.$$

- ▶ A single Poisson solve has no intrinsic marching direction in time.
- ▶ Causal training becomes meaningful only when Module 2 is embedded in an ordered sequence such as dose, irradiation time, defect amplitude, boundary loading, or coupled-iteration continuation.
- ▶ The ordered coordinate is denoted τ or α :

$$\rho(\mathbf{x}; \alpha_0), \rho(\mathbf{x}; \alpha_1), \dots, \rho(\mathbf{x}; \alpha_K), \quad 0 \leq \alpha_0 < \alpha_1 < \dots < \alpha_K \leq 1.$$

- ▶ The CPINN objective is to learn easy or early states before enforcing later, harder, more highly loaded states.

Interpretation: Module 2 CPINN is a causality-respecting curriculum for a family of quasi-static Poisson problems.

Module 2 CPINN: continuation coordinate and staged residuals

A convenient continuation path from easy to full electrostatic loading

$$\rho(\mathbf{x}; \alpha) = (1 - \alpha)\rho_{\text{easy}}(\mathbf{x}) + \alpha\rho_{\text{full}}(\mathbf{x}), \quad 0 \leq \alpha \leq 1$$

Residual at each continuation slice

$$\mathcal{R}_P(\mathbf{x}, \alpha; \theta) = \nabla \cdot (\epsilon \nabla \phi_\theta(\mathbf{x}, \alpha)) + \rho(\mathbf{x}; \alpha).$$

Slice loss

$$\mathcal{L}_k = \frac{1}{N_k} \sum_{m=1}^{N_k} |\mathcal{R}_P(\mathbf{x}_m, \alpha_k; \theta)|^2 + \mathcal{L}_{D,k} + \mathcal{L}_{N,k} + \mathcal{L}_{\text{data},k}.$$

- ▶ $\alpha = 0$ may represent zero defect charge, weak source amplitude, simple boundary loading, or an earlier dose state.
- ▶ $\alpha = 1$ is the full target electrostatic problem.
- ▶ Staged training prevents the network from fitting late/high-damage states before it can satisfy simpler earlier states.

Module 2 CPINN: causal weighting of ordered losses

One causal-PINN strategy weights later residuals by earlier residual success.

$$\mathcal{L}_{\text{causal}} = \sum_{k=0}^K w_k \mathcal{L}_k, \quad w_k = \exp \left[-\eta \sum_{j=0}^{k-1} \mathcal{L}_j \right]$$

- ▶ If early slices have large residuals, then w_k suppresses the influence of later slices.
- ▶ Once early slices are learned, later slices become active because their causal weights increase.
- ▶ This is not a physical modification of Poisson's equation; it is a training schedule imposed on the learning objective.
- ▶ The full loss may be written as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{causal}} + \lambda_D \mathcal{L}_D + \lambda_N \mathcal{L}_N + \lambda_s \mathcal{L}_{\text{data}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}.$$

- ▶ The parameter η controls how aggressively the curriculum enforces earlier states first.

Module 2 CPINN: project use and verification criteria

The CPINN should be accepted only if it improves ordered-state learning without degrading electrostatic fidelity.

- ▶ Compare against the ordinary PINN at every ordered slice:

$$e_{\phi}(\alpha_k) = \frac{\|\phi_{\theta}(\cdot, \alpha_k) - \phi_{\text{FEM}}(\cdot, \alpha_k)\|_2}{\|\phi_{\text{FEM}}(\cdot, \alpha_k)\|_2 + \phi_{\text{floor}}}.$$

- ▶ Check whether late/high-damage states improve because early/low-damage states were learned first.
- ▶ Verify that boundary errors do not grow at large α .
- ▶ Verify that the Poisson residual decreases across the whole continuation path, not only at the final state.
- ▶ Useful ordered coordinates for this project include dose, defect source amplitude, applied bias, and coupled-iteration loading.

Module 3 cPINN: physical time makes causality intrinsic

Module 3 is the preferred causal-PINN base case because it is a genuine initial-value problem.

$$\boxed{\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) - k_{\text{ann}} C + S, \quad C(\mathbf{x}, 0) = C_0(\mathbf{x})}$$

$$\hat{\mathbf{n}} \cdot D \nabla C = 0 \quad \text{on } \partial\Omega, \quad \hat{C}_\theta = \mathcal{N}_\theta(\hat{x}, \hat{y}, \hat{t}).$$

- ▶ $C_0(\mathbf{x})$ is the Module 1 damage-derived initial field; every later state must remain consistent with this causal anchor.
- ▶ Diffusion redistributes defects, $-k_{\text{ann}} C$ removes them locally, and $S(\mathbf{x}, t)$ represents continuing production.
- ▶ Unlike Module 2 Poisson electrostatics, physical time appears directly in the governing equation: the solution at later time depends on the earlier trajectory.
- ▶ “Causal” changes the optimization curriculum, not the defect-physics equation.

Module 3 cPINN: scaled residual and time slices

Only dimensionless losses may enter the causal exponent.

$$\hat{x} = \frac{x}{L}, \quad \hat{y} = \frac{y}{L}, \quad \hat{t} = \frac{t}{t_{\text{ref}}}, \quad \hat{C} = \frac{C}{C_{\text{ref}}}, \quad \text{Da} = k_{\text{ref}} t_{\text{ref}},$$
$$\hat{\mathcal{R}}_{\theta} = \frac{\partial \hat{C}_{\theta}}{\partial \hat{t}} - \hat{\nabla} \cdot (\hat{D} \hat{\nabla} \hat{C}_{\theta}) + \text{Da} \hat{k} \hat{C}_{\theta} - \hat{S}.$$

$$0 = \hat{t}_0 < \hat{t}_1 < \dots < \hat{t}_{N_t} = 1, \quad \hat{\mathcal{L}}_{r,i} = \frac{1}{N_r} \sum_{j=1}^{N_r} \left| \hat{\mathcal{R}}_{\theta}(\hat{\mathbf{x}}_{r,j}^{(i)}, \hat{t}_i) \right|^2.$$

$$\hat{\mathcal{L}}_{\text{IC}} = \frac{1}{N_0} \sum_{j=1}^{N_0} \left| \hat{C}_{\theta}(\hat{\mathbf{x}}_0^{(j)}, 0) - \hat{C}_0(\hat{\mathbf{x}}_0^{(j)}) \right|^2, \quad \mathcal{L}_0 = \lambda_0 \hat{\mathcal{L}}_{\text{IC}}.$$

- ▶ The ordered slices are physical times and must remain sorted when the causal prefix is constructed.
- ▶ Nondimensionalization makes activation independent of whether concentration is stored in m^{-3} , cm^{-3} , or normalized units.
- ▶ Smooth activations such as $\tanh$ are required because the residual contains first-time and second-space derivatives.

Module 3 cPINN: causal weights activate early times first

The initial condition and PDE residuals form the ordered causal prefix.

$$\mathcal{L}_i = \lambda_r \widehat{\mathcal{L}}_{r,i} \quad (i \geq 1), \quad w_0 = 1, \quad w_i = \exp \left[-\epsilon_c \text{ stopgrad} \left(\sum_{k=0}^{i-1} \mathcal{L}_k \right) \right].$$
$$\mathcal{L}_{\text{total}} = \frac{1}{N_t + 1} \sum_{i=0}^{N_t} w_i \mathcal{L}_i + \frac{\lambda_b}{N_t} \sum_{i=1}^{N_t} \widehat{\mathcal{L}}_{b,i} + \frac{\lambda_s}{N_t} \sum_{i=1}^{N_t} \widehat{\mathcal{L}}_{s,i} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}.$$

- ▶ Large early-time error makes later w_i small; as the prefix losses decrease, an activation front moves toward later time.
- ▶ stopgrad prevents the optimizer from reducing the objective by manipulating the weights themselves.
- ▶ A staged schedule such as $\epsilon_c \in \{10^{-2}, 10^{-1}, 10^0, 10^1, 10^2\}$ progressively sharpens temporal ordering.
- ▶ Full activation may be diagnosed by $\min_i w_i > \delta_{\text{stop}}$, with $\delta_{\text{stop}} \approx 0.99$ as a starting criterion.

Module 3 cPINN: verification and coupling

A causal loss is useful only if the learned trajectory also satisfies independent physics checks.

$$\begin{aligned} D = 0, S = 0 &\Rightarrow C(t) = C_0 e^{-k_{\text{ann}} t}, \\ D = \text{constant}, S = 0 &\Rightarrow C(\mathbf{x}, t) = e^{-k_{\text{ann}} t} C_{\text{diff}}(\mathbf{x}, t), \\ \hat{\mathbf{n}} \cdot D \nabla C = 0, S = 0 &\Rightarrow M(t) = \int_{\Omega} C d\Omega = M(0) e^{-k_{\text{ann}} t}. \end{aligned}$$

- ▶ Compare against the ordinary PINN using the same network, initialization, collocation points, optimizer, data budget, and test grid; initially change only $w_i = 1$ to causal w_i .
- ▶ Report field error, maximum error, residual by time slice, w_i history, inventory balance, symmetry, variance growth, and minimum predicted concentration.
- ▶ Progress from exact pure annealing to Gaussian diffusion, diffusion plus annealing, an imported Geant4-like map, temperature-dependent coefficients, and finally multi-species reactions.
- ▶ The validated output $C(\mathbf{x}, t)$ supplies time-ordered defect charge to Module 2 and defect-dependent coefficients to Modules 4–6.

Module 11 GNN: target physical dependency graph

Module 11 learns directed relations among physical variables, not mesh adjacency or numerical solver order.

$$\mathcal{G}_{\text{phys}} = (\mathcal{V}_{\text{phys}}, \mathcal{E}_{\text{phys}}), \quad X_i \rightarrow X_j \iff X_i \text{ directly enters the law for } X_j.$$

- Nodes include fields, coefficients, sources, and state variables from Modules 2–5:

$$C_j, \rho, \phi, \mathbf{E}, T, n, p, \mu_n, \mu_p, D_n, D_p, \tau_n, \tau_p, R, \mathbf{J}_n, \mathbf{J}_p, Q_J, Q_{\text{rad}}.$$

- Edges are directed physical dependencies: algebraic, constitutive, source, reaction, transport, or delayed dynamical effects.
- The learned graph is allowed to contain cycles because real coupled physics contains feedback.
- The goal is edge discovery, mechanism typing, strength estimation, lag classification, and uncertainty quantification.

Module 11 GNN: typed edges and physical meaning

Each candidate edge carries more information than present/absent.

$$X_i \rightarrow X_j \mapsto (A_{ij}, z_{ij}, s_{ij}, \ell_{ij}, \sigma_{ij})$$

- ▶ A_{ij} : learned edge strength or probability.
- ▶ z_{ij} : mechanism type, such as algebraic, constitutive, source, transport, reaction, derivative, or delayed dynamical edge.
- ▶ s_{ij} : sign or monotonic trend when physically meaningful.
- ▶ ℓ_{ij} : lag class, distinguishing instantaneous relations from delayed responses.
- ▶ σ_{ij} : uncertainty or confidence in the inferred edge.

Examples of target relations

$$\begin{aligned} C_j &\rightarrow \rho_{\text{def}}, \\ C_j &\rightarrow \mu_n, \mu_p, \\ \mathbf{E}, n, p &\rightarrow \mathbf{J}_n, \mathbf{J}_p, \end{aligned}$$

$$\begin{aligned} \rho &\rightarrow \phi, \\ C_j &\rightarrow \tau_n, \tau_p, \\ \mathbf{J} \cdot \mathbf{E} &\rightarrow Q_J, \end{aligned}$$

$$\begin{aligned} \phi &\rightarrow \mathbf{E}, \\ T &\rightarrow D_j, k_{\text{ann},j}, \\ Q_J &\rightarrow T. \end{aligned}$$

Module 11 GNN: message passing with latent adjacency

A variable-level GNN updates each physical node using messages from candidate parent variables.

$$\begin{aligned}\mathbf{m}_{ij}^{(\ell)} &= A_{ij} M_{z_{ij}}\left(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}, \mathbf{g}\right), \\ \bar{\mathbf{m}}_j^{(\ell)} &= \sum_{i \neq j} \mathbf{m}_{ij}^{(\ell)}, \\ \mathbf{h}_j^{(\ell+1)} &= U\left(\mathbf{h}_j^{(\ell)}, \bar{\mathbf{m}}_j^{(\ell)}, \mathbf{g}\right).\end{aligned}$$

- ▶ $\mathbf{h}_j$ is the learned representation of physical variable X_j .
- ▶ $\mathbf{e}_{ij}$ stores candidate-edge metadata such as allowed mechanism class, dimensional compatibility, and intervention labels.
- ▶ $\mathbf{g}$ stores global context such as radiation condition, bias, material identity, temperature scale, and time index.
- ▶ The adjacency variable A_{ij} can be supervised when the synthetic generator knows the true graph, or treated as a latent variable when only trajectories are available.
- ▶ Sparsity is essential: without a penalty or prior, the model may explain everything using too many weak edges.

Module 11 GNN: synthetic trajectories and interventions

Synthetic data must vary both parameter values and graph structure.

$$\mathcal{D}^{(r)} = \left\{ \boldsymbol{\theta}^{(r)}, A^{(r)}, Z^{(r)}, L^{(r)}, X^{(r)}(t), I^{(r)} \right\}.$$

- ▶ $\boldsymbol{\theta}^{(r)}$ contains material, source, transport, trapping, thermal, and boundary parameters.
- ▶ $A^{(r)}$ is the ground-truth physical adjacency used to generate the case.
- ▶ Mechanism switches create graph-family variation:

$$s_{ij} \in \{0, 1\}, \quad \gamma_{ij} \sim \text{log-uniform or bounded random strength}.$$

- ▶ Matched interventions isolate edges by changing one mechanism while keeping the same random seed:

$$z_j = 0, \quad \alpha_{\mu, C} = 0, \quad \alpha_{\tau, C} = 0, \quad \gamma_J = 0, \quad T(\mathbf{x}, t) = T_0.$$

- ▶ Training, validation, and test splits should be by complete physical case, not by time samples from the same trajectory.

Module 11 GNN: from learned graph to physical coupling sequence

The learned graph may contain feedback, so the output is a partial physical ordering, not a single serial algorithm.

$$\mathcal{G}_{\text{phys}} \longrightarrow \{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_K\} \longrightarrow \mathcal{G}_{\text{cond}}.$$

- ▶ Strongly connected components $\mathcal{C}_k$ collect mutually dependent feedback groups.
- ▶ Collapsing each feedback group into one supernode gives the condensation graph $\mathcal{G}_{\text{cond}}$, which is acyclic.
- ▶ A topological ordering of $\mathcal{G}_{\text{cond}}$ gives a physically admissible coupling sequence among influence groups:

$$\mathcal{C}_{\pi(1)} \prec \mathcal{C}_{\pi(2)} \prec \dots \prec \mathcal{C}_{\pi(K)}.$$

- ▶ Instantaneous algebraic constraints, such as $\rho \rightarrow \phi \rightarrow \mathbf{E}$, should be separated from delayed dynamical effects, such as $T \rightarrow D_j \rightarrow C_j(t + \Delta t)$.
- ▶ Success means recovering known synthetic edges, rejecting ablated edges, quantifying edge uncertainty, and identifying regime-dependent feedback groups.

Module 2 superconductivity: electrostatics in hybrid SC devices

The superconducting extension of Module 2 is still electrostatics, but the charge response and boundary conditions change.

$$\Omega = \Omega_{\text{sub}} \cup \Omega_{\text{ox}} \cup \Omega_{\text{sc}} \cup \Omega_{\text{JJ}}, \quad \mathbf{E} = -\nabla\phi$$

$$\nabla \cdot (\varepsilon \nabla \phi) = -\rho_{\text{tot}}, \quad \rho_{\text{tot}} = \rho_{\text{def}} + \rho_{\text{trap}} + \rho_{\text{qp}} + \rho_{\text{ind}} + \rho_{\text{ext}}.$$

- ▶ Substrate and oxide regions store radiation-induced defect charge and trapped charge.
- ▶ Superconducting films screen static electric fields and are often modeled as equipotential islands.
- ▶ Quasiparticle charge imbalance can enter as an additional nonequilibrium charge density.
- ▶ The normal Module 2 source term $q(p - n + N_D^+ - N_A^-) + \rho_{\text{def}}$ is replaced by a hybrid-material charge model.

Module 2 superconductivity: screening and equipotential islands

In a metal or superconductor, static charge perturbations are screened over a very short length scale.

$$\rho_{\text{ind}} \approx -\frac{\varepsilon_{\text{sc}}}{\lambda_{\text{scr}}^2} (\phi - \phi_{\text{sc}}),$$
$$\nabla \cdot (\varepsilon_{\text{sc}} \nabla \phi) = -\rho_{\text{ext}} + \frac{\varepsilon_{\text{sc}}}{\lambda_{\text{scr}}^2} (\phi - \phi_{\text{sc}}).$$

- ▶ λ_{scr} is an electrostatic screening length, conceptually closer to Thomas-Fermi screening than London magnetic penetration.
- ▶ In the thin-film static limit, the numerically simpler model is often

$$\phi|_{\Omega_{\text{sc},i}} = V_i, \quad \mathbf{E}_{\parallel} \approx 0 \quad \text{inside the island.}$$

- ▶ The electrochemical potential is the quantity that equilibrates:

$$\mu_{\text{ec}} = \mu_{\text{chem}} - e\phi.$$

- ▶ Radiation-induced local charge is therefore felt mainly through surface charge, island potential shifts, trap occupation, and offset charge.

Module 2 superconductivity: interfaces, capacitance, and offset charge

The most important device-level output is often not the local field inside the metal, but induced island charge.

$$\begin{aligned}\phi_d &= \phi_{sc} \\ \hat{\mathbf{n}} \cdot (\mathbf{D}_d - \mathbf{D}_{sc}) &= \sigma_{\text{int}}\end{aligned}\quad \begin{array}{l} \text{on ideal electrostatic interfaces,} \\ \text{with interface trapped charge.} \end{array}$$

$$\mathbf{Q} = \mathbf{C}\mathbf{V} + \mathbf{Q}_{\text{off}}, \quad Q_{\text{off},i} = \int_{\Omega_d} \rho_{\text{rad}}(\mathbf{x}) \psi_i(\mathbf{x}) d\Omega$$

- ▶ $\mathbf{C}$ is the capacitance matrix for the superconducting island network.
- ▶ ψ_i is a weighting potential: solve Laplace's equation with island i at unit voltage and other conductors grounded.
- ▶ Radiation-generated defects and trapped charges shift $\mathbf{Q}_{\text{off}}$, which perturbs charge-sensitive superconducting devices.
- ▶ For transmon-like devices, offset charge is partially suppressed but still relevant through traps, quasiparticles, and dielectric loss pathways.

Module 2 superconductivity: minimal equations to carry forward

The first superconducting Module 2 implementation should not overclaim microscopic condensate dynamics.

$$\begin{aligned}\nabla \cdot (\varepsilon \nabla \phi) &= -(\rho_{\text{def}} + \rho_{\text{trap}} + \rho_{\text{qp}}) && \text{in dielectric/substrate regions,} \\ \phi &= V_i && \text{on superconducting island } i, \\ \mathbf{Q} &= \mathbf{C}\mathbf{V} + \mathbf{Q}_{\text{off}}, \quad \mathbf{E} = -\nabla \phi.\end{aligned}$$

- ▶ Use superconducting pads as electrostatic conductors first; add finite screening only when needed.
- ▶ Track oxide/interface traps separately from bulk substrate defect charge.
- ▶ Treat quasiparticle charge imbalance as a nonequilibrium correction, not as ordinary semiconductor carrier density.
- ▶ Outputs to later modules: ϕ , $\mathbf{E}$, surface/interface fields, $\mathbf{Q}_{\text{off}}$, capacitance sensitivity, and trap-field coupling.

Module 3 superconductivity: structural defects are not quasiparticles

Module 3 remains the slow material-state module: structural disorder, traps, TLSs, and junction-barrier defects.

$$\mathbf{C}_{\text{SC}} = \{C_i^{\text{sub}}, C_i^{\text{ox}}, C_i^{\text{film}}, N_t, f_t, N_{\text{TLS}}, C_i^{\text{JJ}}\}$$

- ▶ Cryogenic operation separates slow structural variables from fast excitations.
- ▶ Structural defects and traps can persist for long times; quasiparticles and pair-breaking phonons are fast auxiliary variables handled mainly by Modules 4–5.
- ▶ The same radiation event can create substrate damage, oxide trapped charge, interface TLS changes, superconducting-film disorder, and junction-barrier defects.
- ▶ The project should avoid merging all of these into one warm-semiconductor defect concentration.

Module 3 superconductivity: material-resolved diffusion-reaction

Each material region can carry its own defect species, mobility, reaction network, and boundary behavior.

$$\frac{\partial C_i^{(m)}}{\partial t} = \nabla \cdot \left(D_i^{(m)}(T) \nabla C_i^{(m)} \right) + R_i^{(m)}(\mathbf{C}^{(m)}, T, n, p) + S_i^{(m)}$$

$$D_i^{(m)}(T) = D_{i0}^{(m)} \exp \left(-\frac{E_{m,i}^{(m)}}{k_B T} \right),$$

$$k_{\text{ann},i}^{(m)}(T) = k_{i0}^{(m)} \exp \left(-\frac{E_{\text{ann},i}^{(m)}}{k_B T} \right).$$

- ▶ At millikelvin base temperature, thermally activated diffusion can be nearly frozen.
- ▶ Radiation can still produce transient local hot spots, nonthermal rearrangement, charge-state changes, and trap occupation changes.
- ▶ Interface conditions may allow reflection, trapping, sink behavior, or defect conversion at material boundaries.

Module 3 superconductivity: traps, TLSs, and junction defects

Interface and junction defects are often more device-relevant than bulk displaced atoms.

$$\rho_{\text{trap}}(\mathbf{x}, t) = q \sum_a z_a N_{t,a}(\mathbf{x}) f_a(\mathbf{x}, t),$$

$$\frac{\partial f_a}{\partial t} = c_a^+ (1 - f_a) - c_a^- f_a,$$

$$P_{\text{TLS}}(E) \rightarrow \{\tan \delta, 1/f \text{ charge noise, qubit frequency noise}\}.$$

- ▶ N_t is a trap density; f_t is trap occupation, which controls charge state and electric-field noise.
- ▶ TLSs are modeled as localized electric dipoles in a heterogeneous polarizability landscape; they couple to microwave electric fields and can produce dielectric loss or fluctuating offset charge.
- ▶ Josephson-junction barrier defects can affect critical current, tunneling asymmetry, subgap conductance, and switching noise.
- ▶ Module 3 should pass trap/TLS/barrier state variables to Modules 2, 5, 6, 7, and 8.

Module 3 superconductivity: coupling outputs and first implementation

The first useful implementation is a region-labeled structural-state update, not a full microscopic defect simulation.

$$\mathbf{C}_{\text{out}} \longrightarrow \{\rho_{\text{def}}, \rho_{\text{trap}}, Q_{\text{off}}, N_{\text{TLS}}, \delta I_C, \Gamma_{\text{trap}}, \tan \delta\}$$

- ▶ To Module 2: charged structural defects and trap occupation produce electrostatic source terms and offset charge.
- ▶ To Module 4: defect formation, relaxation, and trapping can exchange energy with phonon populations.
- ▶ To Module 5: traps and disorder modify quasiparticle trapping, poisoning, tunneling, and recombination channels.
- ▶ Verification targets:

$$C_i(t) = C_i(0)e^{-k_i t} \quad (D_i = 0), \quad \int_{\Omega_m} C_i d\Omega = \text{constant} \quad (R_i = S_i = 0).$$

- ▶ State variables should be grouped by material: substrate, oxide/interface, superconducting film, and junction barrier.

Module 4 superconductivity: one temperature field is insufficient

At cryogenic temperatures, deposited radiation energy partitions among several weakly equilibrated channels.

$$E_{\text{dep}} \rightarrow \{U_{\text{sub phonon}}, U_{\text{pb phonon}}, U_{\text{qp}}, U_{\text{trap}}, U_{\text{escape}}\}$$

- ▶ A single warm-device heat equation cannot distinguish harmless low-energy phonons from pair-breaking phonons.
- ▶ Superconducting films have exponentially suppressed electronic heat capacity below T_c .
- ▶ Interfaces can bottleneck energy flow through Kapitza resistance.
- ▶ Module 4 should represent substrate phonons, pair-breaking phonons, quasiparticle energy, and escape to the package or heat sink at a reduced-model level.

Module 4 superconductivity: pair-breaking threshold and energy partition

The superconducting gap creates a threshold for converting phonons into quasiparticles.

$$\hbar\omega \geq 2\Delta \quad \Rightarrow \quad \text{pair-breaking phonon can create two quasiparticles}$$

$$\dot{U}_{\text{pb}} = \eta_{\text{pb}} \dot{E}_{\text{dep}} - P_{\text{pair}} - P_{\text{esc}} - P_{\text{down}},$$

$$\dot{U}_{\text{ph}} = (1 - \eta_{\text{pb}}) \dot{E}_{\text{dep}} + P_{\text{down}} - P_{\text{bath}}.$$

- ▶ η_{pb} is an effective fraction of deposited energy placed into pair-breaking-capable phonons.
- ▶ Downconversion moves high-energy phonons below the pair-breaking threshold.
- ▶ Escape terms represent phonon loss to substrate, package, backside metal, or external bath.
- ▶ The relevant output is not only T ; it is also the time-integrated pair-breaking energy exposure.

Module 4 superconductivity: quasiparticle-phonon kinetics

A reduced Rothwarf-Taylor structure couples quasiparticle density and pair-breaking phonon population.

$$\begin{aligned}\frac{dn_{\text{qp}}}{dt} &= 2\Gamma_{\text{pb}}N_{\text{pb}} - R_{\text{qp}}n_{\text{qp}}^2 - \Gamma_{\text{trap}}n_{\text{qp}} + G_{\text{inj}}, \\ \frac{dN_{\text{pb}}}{dt} &= -\Gamma_{\text{pb}}N_{\text{pb}} + \frac{1}{2}R_{\text{qp}}n_{\text{qp}}^2 - \Gamma_{\text{esc}}N_{\text{pb}} + S_{\text{pb}}.\end{aligned}$$

- ▶ Pair-breaking phonons increase n_{qp} .
- ▶ Quasiparticle recombination emits high-energy phonons, which can break more pairs if they do not escape.
- ▶ Traps and normal-metal quasiparticle sinks remove quasiparticles from active superconducting regions.
- ▶ Module 4 supplies N_{pb} , phonon energy, and thermal boundary losses to Module 5 and Module 6.

Module 4 superconductivity: boundaries, hierarchy, and metrics

The practical path is a hierarchy from scalar thermal diffusion to reduced spectral phonon/QP energetics.

$$\begin{aligned} -k_1 \nabla T_1 \cdot \hat{\mathbf{n}} &= G_K (T_1 - T_2) && \text{Kapitza/interface boundary,} \\ C_{\text{ph}} \frac{\partial T_{\text{ph}}}{\partial t} &= \nabla \cdot (k_{\text{ph}} \nabla T_{\text{ph}}) + P_{\text{dep}} - P_{\text{esc}} - P_{\text{pair}}. \end{aligned}$$

- ▶ Level 0: ordinary thermal diffusion with cryogenic material constants.
- ▶ Level 1: add Kapitza boundary resistance and package/bath coupling.
- ▶ Level 2: track pair-breaking phonon population or energy.
- ▶ Level 3: couple pair-breaking phonons to quasiparticle generation/recombination.
- ▶ Useful outputs: peak temperature, thermal relaxation time, escaped power, pair-breaking fluence, and QP-generation source for Module 5.

Module 5 superconductivity: carrier response changes below T_c

Ordinary drift-diffusion is no longer the dominant transport model in superconducting films.

$$\mathbf{J}_{\text{tot}} = \mathbf{J}_s + \mathbf{J}_{\text{qp}} + \mathbf{J}_{\text{leak}} + \mathbf{J}_{\text{disp}}$$

- ▶ Semiconductor electrons and holes may still matter in substrates, dielectrics, or radiation bursts.
- ▶ In superconducting films, the condensate carries dissipationless supercurrent and quasiparticles carry dissipative nonequilibrium current.
- ▶ Below T_c , the relevant mobile excitations are Cooper-pair condensate phase and quasiparticle population, not ordinary n, p drift-diffusion.
- ▶ Module 5 therefore becomes a hybrid carrier-response module: substrate charge, trap occupation, quasiparticles, Josephson transport, and device-response metrics.

Module 5 superconductivity: condensate and Josephson transport

Supercurrent is governed by phase, not by Ohmic mobility.

$$\mathbf{J}_s = \frac{2e\hbar}{m^*} |\Psi|^2 \left(\nabla\theta - \frac{2e}{\hbar} \mathbf{A} \right),$$

$$I_s = I_c \sin \delta,$$

$$V = \frac{\Phi_0}{2\pi} \frac{d\delta}{dt}, \quad \Phi_0 = \frac{h}{2e}.$$

- ▶ θ is condensate phase; δ is the gauge-invariant phase difference across a Josephson junction.
- ▶ Radiation-induced junction-barrier defects can shift I_c , subgap leakage, and tunneling asymmetry.
- ▶ Capacitive island dynamics enter through charge variables rather than ordinary drift current across the barrier.
- ▶ In the lowest-fidelity model, Josephson elements are compact circuit relations coupled to field-derived offset charge and QP rates.

Module 5 superconductivity: quasiparticle diffusion, trapping, and poisoning

Quasiparticles are the main dissipative carrier population inside superconducting films.

$$\frac{\partial n_{\text{qp}}}{\partial t} = \nabla \cdot (D_{\text{qp}} \nabla n_{\text{qp}}) + G_{\text{qp}} - R_{\text{qp}} n_{\text{qp}}^2 - \frac{n_{\text{qp}}}{\tau_{\text{trap}}} - \frac{n_{\text{qp}}}{\tau_{\text{esc}}}$$

$$\mathbf{J}_{\text{qp}} \approx -eD_{\text{qp}} \nabla n_{\text{qp}} + \sigma_{\text{qp}} (-\nabla \phi),$$

$$\Gamma_1^{\text{qp}} \propto x_{\text{qp}}, \quad x_{\text{qp}} \equiv \frac{n_{\text{qp}}}{2N_0\Delta}.$$

- ▶ Pair-breaking phonons from Module 4 provide G_{qp} .
- ▶ Normal-metal traps and gap-engineered regions reduce active quasiparticle density.
- ▶ QP tunneling across a junction can produce poisoning, relaxation, parity switching, and excess loss.

Module 5 superconductivity: offset charge and device metrics

Radiation-generated charge and traps couple to superconducting islands through capacitance and weighting fields.

$$\begin{aligned} \mathbf{Q} &= \mathbf{C}\mathbf{V} + \mathbf{Q}_{\text{off}}, \\ n_g &= \frac{Q_{\text{off}}}{2e}, \\ Q_{\text{off},i} &= \int \rho_{\text{trap}}(\mathbf{x}, t) \psi_i(\mathbf{x}) d\Omega. \end{aligned}$$

- ▶ n_g is the dimensionless offset charge relevant to island-based superconducting circuits.
- ▶ Defects alter device response through Q_{off} , I_c , dielectric loss, QP density, and QP tunneling rates.
- ▶ Useful Module 5 outputs include:

$$n_{\text{qp}}, \quad x_{\text{qp}}, \quad \Gamma_1^{\text{qp}}, \quad \Gamma_{\text{poison}}, \quad \delta I_c, \quad \delta n_g.$$

- ▶ These are device-degradation variables, not merely carrier-density maps.

Module 6 superconductivity: expanded coupled state vector

The superconducting coupling layer must separate electrostatics, structural defects, phonons, quasiparticles, and device response.

$$\mathbf{u}_{\text{SC}} = \{\phi, \mathbf{E}, Q_{\text{off}}, \mathbf{C}_{\text{def}}, N_t, f_t, N_{\text{TLS}}, T_{\text{ph}}, N_{\text{pb}}, n_{\text{qp}}, \delta, I_c, \Gamma_1\}$$

- ▶ Module 2 supplies electrostatics, island potentials, capacitance response, and offset charge.
- ▶ Module 3 supplies structural defects, traps, TLSs, barrier disorder, and disorder-driven parameter shifts.
- ▶ Module 4 supplies substrate phonon energy, pair-breaking phonons, escape, and thermal relaxation.
- ▶ Module 5 supplies quasiparticles, Josephson response, QP tunneling, poisoning, and device-rate metrics.

Module 6 superconductivity: radiation-event partition and residuals

A radiation event must be partitioned into charge, structural damage, phonons, and superconducting excitations.

$$E_{\text{dep}} \rightarrow \{S_{\text{def}}, \rho_{\text{ion}}, S_{\text{trap}}, S_{\text{pb}}, G_{\text{qp}}, P_{\text{heat}}\}$$

$$\mathcal{R}_2 : \quad \nabla \cdot (\varepsilon \nabla \phi) + \rho_{\text{def}} + \rho_{\text{trap}} + \rho_{\text{qp}} = 0,$$

$$\mathcal{R}_3 : \quad \partial_t \mathbf{C}_{\text{def}} - \mathcal{F}_{\text{def}}(\mathbf{C}_{\text{def}}, T_{\text{ph}}, f_t) = 0,$$

$$\mathcal{R}_4 : \quad \partial_t \{T_{\text{ph}}, N_{\text{pb}}\} - \mathcal{F}_{\text{ph}}(E_{\text{dep}}, n_{\text{qp}}) = 0,$$

$$\mathcal{R}_5 : \quad \partial_t n_{\text{qp}} - \mathcal{F}_{\text{qp}}(N_{\text{pb}}, \mathbf{C}_{\text{def}}, \text{JJ}) = 0.$$

Module 6 superconductivity: feedback loops and time-scale separation

The superconducting coupling graph contains fast excitation loops and slow material-state loops.

$$\begin{aligned} E_{\text{dep}} &\rightarrow N_{\text{pb}} \rightarrow n_{\text{qp}} \rightarrow \Gamma_1, \Gamma_{\text{poison}}, \\ n_{\text{qp}} &\rightarrow \text{recombination phonons} \rightarrow N_{\text{pb}}, \\ C_{\text{def}}, f_t &\rightarrow \rho_{\text{trap}}, Q_{\text{off}} \rightarrow \delta n_g, \\ C_{\text{JJ}} &\rightarrow I_c, \Gamma_{\text{qp}} \rightarrow \text{device response}. \end{aligned}$$

- ▶ Fast variables: ionization charge, pair-breaking phonons, quasiparticles, junction tunneling rates.
- ▶ Intermediate variables: trap occupation, QP trapping, phonon escape, temperature relaxation.
- ▶ Slow variables: structural defects, interface disorder, TLS population, junction-barrier degradation.
- ▶ Coupling must respect time scales; forcing all variables into one explicit update can create artificial physics.

Module 6 superconductivity: consistency checks and first algorithm

The first implementation should prioritize conservation and correct field-transfer bookkeeping.

$$E_{\text{dep}} \approx U_{\text{ph}} + U_{\text{pb}} + U_{\text{qp}} + U_{\text{trap}} + E_{\text{esc}} + E_{\text{loss}},$$
$$Q_{\text{off},i} = \int \rho_{\text{trap}} \psi_i d\Omega, \quad \rho_{\text{tot}} = \rho_{\text{def}} + \rho_{\text{trap}} + \rho_{\text{qp}}.$$

1. Partition each radiation event into structural damage, trapped charge, pair-breaking phonons, heat, and QP sources.
2. Update slow structural/trap state only on appropriate time scales.
3. Solve electrostatic offset charge from the current trap/defect/QP charge state.
4. Update phonon-QP kinetics and pass rates to device-response metrics.
5. Verify energy partition, charge consistency, zero-source equilibrium, and one-way-coupling limits.